MSG No. 22.48 F_S222 [Type IV, orthorhombic]

Table 1: Wyckoff site: `8a`, site symmetry: `222`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4]`
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[9,10,11,12]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[13,14,15,16]`
5	$\left[0, 0, \frac{1}{2}\right]$	`[17,18,19,20]`
6	$\left[0, \frac{1}{2}, 0\right]$	`[21,22,23,24]`
7	$\left[\frac{1}{2}, 0, 0\right]$	`[25,26,27,28]`
8	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[29,30,31,32]`

Table 2: Wyckoff site: `8b`, site symmetry: `22'2'`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, 0\right]$	`[1,2,27,28]`
2	$\left[\frac{3}{4}, 0, 0\right]$	`[3,4,25,26]`
3	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6,31,32]`
4	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[7,8,29,30]`
5	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[9,10,19,20]`
6	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[11,12,17,18]`
7	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[13,14,23,24]`
8	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[15,16,21,22]`

Table 3: Wyckoff site: `8c`, site symmetry: `2'22'`

No.	position	mapping
1	$\left[0, \frac{1}{4}, 0\right]$	`[1,3,22,24]`
2	$\left[0, \frac{3}{4}, 0\right]$	`[2,4,21,23]`
3	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[5,7,18,20]`
4	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[6,8,17,19]`
5	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}\right]$	`[9,11,30,32]`
6	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{2}\right]$	`[10,12,29,31]`
7	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[13,15,26,28]`
8	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[14,16,25,27]`

Table 4: Wyckoff site: `8d`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,4,18,19]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[2,3,17,20]`
3	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[5,8,22,23]`
4	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[6,7,21,24]`
5	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[9,12,26,27]`

continued ...

Table 4

No.	position	mapping
6	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[10,11,25,28]
7	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	[13,16,30,31]
8	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[14,15,29,32]

Table 5: Wyckoff site: 8e, site symmetry: 2'2'2

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	[1,16,22,27]
2	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[2,15,21,28]
3	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[3,14,24,25]
4	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	[4,13,23,26]
5	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[5,12,18,31]
6	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[6,11,17,32]
7	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[7,10,20,29]
8	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[8,9,19,30]

Table 6: Wyckoff site: 8f, site symmetry: 2'22'

No.	position	mapping
1	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	[1,11,18,28]
2	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	[2,12,17,27]
3	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	[3,9,20,26]
4	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	[4,10,19,25]
5	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[5,15,22,32]
6	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[6,16,21,31]
7	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[7,13,24,30]
8	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[8,14,23,29]

Table 7: Wyckoff site: 8g, site symmetry: 22'2'

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	[1,6,19,24]
2	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	[2,5,20,23]
3	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	[3,8,17,22]
4	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[4,7,18,21]
5	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[9,14,27,32]
6	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[10,13,28,31]
7	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	[11,16,25,30]
8	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[12,15,26,29]

Table 8: Wyckoff site: `8h`, site symmetry: `222`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,6,11,16]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[2,5,12,15]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,8,9,14]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[4,7,10,13]`
5	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[17,22,27,32]`
6	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[18,21,28,31]`
7	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[19,24,25,30]`
8	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[20,23,26,29]`

Table 9: Wyckoff site: `16i`, site symmetry: `2..`

No.	position	mapping
1	$[x, 0, 0]$	`[1,2]`
2	$[-x, 0, 0]$	`[3,4]`
3	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6]`
4	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[7,8]`
5	$\left[x+\frac{1}{2}, 0, \frac{1}{2}\right]$	`[9,10]`
6	$\left[\frac{1}{2}-x, 0, \frac{1}{2}\right]$	`[11,12]`
7	$\left[x+\frac{1}{2}, \frac{1}{2}, 0\right]$	`[13,14]`
8	$\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$	`[15,16]`
9	$\left[x, 0, \frac{1}{2}\right]$	`[17,18]`
10	$\left[-x, 0, \frac{1}{2}\right]$	`[19,20]`
11	$\left[x, \frac{1}{2}, 0\right]$	`[21,22]`
12	$\left[-x, \frac{1}{2}, 0\right]$	`[23,24]`
13	$\left[x+\frac{1}{2}, 0, 0\right]$	`[25,26]`
14	$\left[\frac{1}{2}-x, 0, 0\right]$	`[27,28]`
15	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[29,30]`
16	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$	`[31,32]`

Table 10: Wyckoff site: `16j`, site symmetry: `2'..`

No.	position	mapping
1	$\left[x, 0, \frac{1}{4}\right]$	`[1,18]`
2	$\left[x, 0, \frac{3}{4}\right]$	`[2,17]`
3	$\left[-x, 0, \frac{3}{4}\right]$	`[3,20]`
4	$\left[-x, 0, \frac{1}{4}\right]$	`[4,19]`
5	$\left[x, \frac{1}{2}, \frac{3}{4}\right]$	`[5,22]`
6	$\left[x, \frac{1}{2}, \frac{1}{4}\right]$	`[6,21]`
7	$\left[-x, \frac{1}{2}, \frac{1}{4}\right]$	`[7,24]`
8	$\left[-x, \frac{1}{2}, \frac{3}{4}\right]$	`[8,23]`
9	$\left[x+\frac{1}{2}, 0, \frac{3}{4}\right]$	`[9,26]`
10	$\left[x+\frac{1}{2}, 0, \frac{1}{4}\right]$	`[10,25]`
11	$\left[\frac{1}{2}-x, 0, \frac{1}{4}\right]$	`[11,28]`
12	$\left[\frac{1}{2}-x, 0, \frac{3}{4}\right]$	`[12,27]`

continued ...

Table 10

No.	position	mapping
13	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[13,30]
14	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[14,29]
15	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[15,32]
16	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[16,31]

Table 11: Wyckoff site: 16k, site symmetry: 2'..

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ 0\right]$	[1,22]
2	$\left[x,\ \frac{3}{4},\ 0\right]$	[2,21]
3	$\left[-x,\ \frac{1}{4},\ 0\right]$	[3,24]
4	$\left[-x,\ \frac{3}{4},\ 0\right]$	[4,23]
5	$\left[x,\ \frac{3}{4},\ \frac{1}{2}\right]$	[5,18]
6	$\left[x,\ \frac{1}{4},\ \frac{1}{2}\right]$	[6,17]
7	$\left[-x,\ \frac{3}{4},\ \frac{1}{2}\right]$	[7,20]
8	$\left[-x,\ \frac{1}{4},\ \frac{1}{2}\right]$	[8,19]
9	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}\right]$	[9,30]
10	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$	[10,29]
11	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{2}\right]$	[11,32]
12	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{2}\right]$	[12,31]
13	$\left[x+\frac{1}{2},\ \frac{3}{4},\ 0\right]$	[13,26]
14	$\left[x+\frac{1}{2},\ \frac{1}{4},\ 0\right]$	[14,25]
15	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ 0\right]$	[15,28]
16	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ 0\right]$	[16,27]

Table 12: Wyckoff site: 16l, site symmetry: 2..

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	[1,6]
2	$\left[x,\ \frac{3}{4},\ \frac{3}{4}\right]$	[2,5]
3	$\left[-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[3,8]
4	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[4,7]
5	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{3}{4}\right]$	[9,14]
6	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}\right]$	[10,13]
7	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	[11,16]
8	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	[12,15]
9	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[17,22]
10	$\left[x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[18,21]
11	$\left[-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	[19,24]
12	$\left[-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	[20,23]
13	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{4}\right]$	[25,30]
14	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}\right]$	[26,29]
15	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[27,32]
16	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[28,31]

Table 13: Wyckoff site: `16m`, site symmetry: `.2.`

No.	position	mapping
1	$[0,\ y,\ 0]$	`[1,3]`
2	$[0,\ -y,\ 0]$	`[2,4]`
3	$\left[0,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[5,7]`
4	$\left[0,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[6,8]`
5	$\left[\frac{1}{2},\ y,\ \frac{1}{2}\right]$	`[9,11]`
6	$\left[\frac{1}{2},\ -y,\ \frac{1}{2}\right]$	`[10,12]`
7	$\left[\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	`[13,15]`
8	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	`[14,16]`
9	$\left[0,\ y,\ \frac{1}{2}\right]$	`[17,19]`
10	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[18,20]`
11	$\left[0,\ y+\frac{1}{2},\ 0\right]$	`[21,23]`
12	$\left[0,\ \frac{1}{2}-y,\ 0\right]$	`[22,24]`
13	$\left[\frac{1}{2},\ y,\ 0\right]$	`[25,27]`
14	$\left[\frac{1}{2},\ -y,\ 0\right]$	`[26,28]`
15	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[29,31]`
16	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[30,32]`

Table 14: Wyckoff site: `16n`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{4}\right]$	`[1,19]`
2	$\left[0,\ -y,\ \frac{3}{4}\right]$	`[2,20]`
3	$\left[0,\ y,\ \frac{3}{4}\right]$	`[3,17]`
4	$\left[0,\ -y,\ \frac{1}{4}\right]$	`[4,18]`
5	$\left[0,\ y+\frac{1}{2},\ \frac{3}{4}\right]$	`[5,23]`
6	$\left[0,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	`[6,24]`
7	$\left[0,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[7,21]`
8	$\left[0,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[8,22]`
9	$\left[\frac{1}{2},\ y,\ \frac{3}{4}\right]$	`[9,27]`
10	$\left[\frac{1}{2},\ -y,\ \frac{1}{4}\right]$	`[10,28]`
11	$\left[\frac{1}{2},\ y,\ \frac{1}{4}\right]$	`[11,25]`
12	$\left[\frac{1}{2},\ -y,\ \frac{3}{4}\right]$	`[12,26]`
13	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[13,31]`
14	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[14,32]`
15	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{3}{4}\right]$	`[15,29]`
16	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	`[16,30]`

Table 15: Wyckoff site: `16o`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ 0\right]$	`[1,27]`
2	$\left[\frac{1}{4},\ -y,\ 0\right]$	`[2,28]`
3	$\left[\frac{3}{4},\ y,\ 0\right]$	`[3,25]`
4	$\left[\frac{3}{4},\ -y,\ 0\right]$	`[4,26]`

continued ...

Table 15

No.	position	mapping
5	$\left[\frac{1}{4},\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[5,31]
6	$\left[\frac{1}{4},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[6,32]
7	$\left[\frac{3}{4},\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[7,29]
8	$\left[\frac{3}{4},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[8,30]
9	$\left[\frac{3}{4},\, y,\, \frac{1}{2}\right]$	[9,19]
10	$\left[\frac{3}{4},\, -y,\, \frac{1}{2}\right]$	[10,20]
11	$\left[\frac{1}{4},\, y,\, \frac{1}{2}\right]$	[11,17]
12	$\left[\frac{1}{4},\, -y,\, \frac{1}{2}\right]$	[12,18]
13	$\left[\frac{3}{4},\, y+\frac{1}{2},\, 0\right]$	[13,23]
14	$\left[\frac{3}{4},\, \frac{1}{2}-y,\, 0\right]$	[14,24]
15	$\left[\frac{1}{4},\, y+\frac{1}{2},\, 0\right]$	[15,21]
16	$\left[\frac{1}{4},\, \frac{1}{2}-y,\, 0\right]$	[16,22]

Table 16: Wyckoff site: 16p, site symmetry: .2.

No.	position	mapping
1	$\left[\frac{1}{4},\, y,\, \frac{1}{4}\right]$	[1,11]
2	$\left[\frac{1}{4},\, -y,\, \frac{3}{4}\right]$	[2,12]
3	$\left[\frac{3}{4},\, y,\, \frac{3}{4}\right]$	[3,9]
4	$\left[\frac{3}{4},\, -y,\, \frac{1}{4}\right]$	[4,10]
5	$\left[\frac{1}{4},\, y+\frac{1}{2},\, \frac{3}{4}\right]$	[5,15]
6	$\left[\frac{1}{4},\, \frac{1}{2}-y,\, \frac{1}{4}\right]$	[6,16]
7	$\left[\frac{3}{4},\, y+\frac{1}{2},\, \frac{1}{4}\right]$	[7,13]
8	$\left[\frac{3}{4},\, \frac{1}{2}-y,\, \frac{3}{4}\right]$	[8,14]
9	$\left[\frac{1}{4},\, y,\, \frac{3}{4}\right]$	[17,27]
10	$\left[\frac{1}{4},\, -y,\, \frac{1}{4}\right]$	[18,28]
11	$\left[\frac{3}{4},\, y,\, \frac{1}{4}\right]$	[19,25]
12	$\left[\frac{3}{4},\, -y,\, \frac{3}{4}\right]$	[20,26]
13	$\left[\frac{1}{4},\, y+\frac{1}{2},\, \frac{1}{4}\right]$	[21,31]
14	$\left[\frac{1}{4},\, \frac{1}{2}-y,\, \frac{3}{4}\right]$	[22,32]
15	$\left[\frac{3}{4},\, y+\frac{1}{2},\, \frac{3}{4}\right]$	[23,29]
16	$\left[\frac{3}{4},\, \frac{1}{2}-y,\, \frac{1}{4}\right]$	[24,30]

Table 17: Wyckoff site: 16q, site symmetry: ..2

No.	position	mapping
1	$[0,\, 0,\, z]$	[1,4]
2	$[0,\, 0,\, -z]$	[2,3]
3	$\left[0,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	[5,8]
4	$\left[0,\, \frac{1}{2},\, \frac{1}{2}-z\right]$	[6,7]
5	$\left[\frac{1}{2},\, 0,\, z+\frac{1}{2}\right]$	[9,12]
6	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}-z\right]$	[10,11]
7	$\left[\frac{1}{2},\, \frac{1}{2},\, z\right]$	[13,16]
8	$\left[\frac{1}{2},\, \frac{1}{2},\, -z\right]$	[14,15]

continued ...

Table 17

No.	position	mapping
9	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	[17,20]
10	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	[18,19]
11	$\left[0,\ \frac{1}{2},\ z\right]$	[21,24]
12	$\left[0,\ \frac{1}{2},\ -z\right]$	[22,23]
13	$\left[\frac{1}{2},\ 0,\ z\right]$	[25,28]
14	$\left[\frac{1}{2},\ 0,\ -z\right]$	[26,27]
15	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[29,32]
16	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[30,31]

Table 18: Wyckoff site: 16r, site symmetry: ..2'

No.	position	mapping
1	$\left[\frac{1}{4},\ 0,\ z\right]$	[1,28]
2	$\left[\frac{1}{4},\ 0,\ -z\right]$	[2,27]
3	$\left[\frac{3}{4},\ 0,\ -z\right]$	[3,26]
4	$\left[\frac{3}{4},\ 0,\ z\right]$	[4,25]
5	$\left[\frac{1}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[5,32]
6	$\left[\frac{1}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[6,31]
7	$\left[\frac{3}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[7,30]
8	$\left[\frac{3}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[8,29]
9	$\left[\frac{3}{4},\ 0,\ z+\frac{1}{2}\right]$	[9,20]
10	$\left[\frac{3}{4},\ 0,\ \frac{1}{2}-z\right]$	[10,19]
11	$\left[\frac{1}{4},\ 0,\ \frac{1}{2}-z\right]$	[11,18]
12	$\left[\frac{1}{4},\ 0,\ z+\frac{1}{2}\right]$	[12,17]
13	$\left[\frac{3}{4},\ \frac{1}{2},\ z\right]$	[13,24]
14	$\left[\frac{3}{4},\ \frac{1}{2},\ -z\right]$	[14,23]
15	$\left[\frac{1}{4},\ \frac{1}{2},\ -z\right]$	[15,22]
16	$\left[\frac{1}{4},\ \frac{1}{2},\ z\right]$	[16,21]

Table 19: Wyckoff site: 16s, site symmetry: ..2'

No.	position	mapping
1	$\left[0,\ \frac{1}{4},\ z\right]$	[1,24]
2	$\left[0,\ \frac{3}{4},\ -z\right]$	[2,23]
3	$\left[0,\ \frac{1}{4},\ -z\right]$	[3,22]
4	$\left[0,\ \frac{3}{4},\ z\right]$	[4,21]
5	$\left[0,\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[5,20]
6	$\left[0,\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[6,19]
7	$\left[0,\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[7,18]
8	$\left[0,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[8,17]
9	$\left[\frac{1}{2},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[9,32]
10	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[10,31]
11	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[11,30]
12	$\left[\frac{1}{2},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[12,29]

continued ...

Table 19

No.	position	mapping
13	$\left[\frac{1}{2}, \frac{3}{4}, z\right]$	[13,28]
14	$\left[\frac{1}{2}, \frac{1}{4}, -z\right]$	[14,27]
15	$\left[\frac{1}{2}, \frac{3}{4}, -z\right]$	[15,26]
16	$\left[\frac{1}{2}, \frac{1}{4}, z\right]$	[16,25]

Table 20: Wyckoff site: 16t, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	[1,16]
2	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	[2,15]
3	$\left[\frac{3}{4}, \frac{1}{4}, -z\right]$	[3,14]
4	$\left[\frac{3}{4}, \frac{3}{4}, z\right]$	[4,13]
5	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	[5,12]
6	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	[6,11]
7	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	[7,10]
8	$\left[\frac{3}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	[8,9]
9	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	[17,32]
10	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	[18,31]
11	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	[19,30]
12	$\left[\frac{3}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	[20,29]
13	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	[21,28]
14	$\left[\frac{1}{4}, \frac{1}{4}, -z\right]$	[22,27]
15	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	[23,26]
16	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	[24,25]

Table 21: Wyckoff site: 32u, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$[x, -y, -z]$	[2]
3	$[-x, y, -z]$	[3]
4	$[-x, -y, z]$	[4]
5	$\left[x, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[5]
6	$\left[x, \frac{1}{2}-y, \frac{1}{2}-z\right]$	[6]
7	$\left[-x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[7]
8	$\left[-x, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[8]
9	$\left[x+\frac{1}{2}, y, z+\frac{1}{2}\right]$	[9]
10	$\left[x+\frac{1}{2}, -y, \frac{1}{2}-z\right]$	[10]
11	$\left[\frac{1}{2}-x, y, \frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-x, -y, z+\frac{1}{2}\right]$	[12]
13	$\left[x+\frac{1}{2}, y+\frac{1}{2}, z\right]$	[13]
14	$\left[x+\frac{1}{2}, \frac{1}{2}-y, -z\right]$	[14]
15	$\left[\frac{1}{2}-x, y+\frac{1}{2}, -z\right]$	[15]
16	$\left[\frac{1}{2}-x, \frac{1}{2}-y, z\right]$	[16]

continued ...

Table 21

No.	position	mapping
17	$\left[x,\, y,\, z+\frac{1}{2}\right]$	[17]
18	$\left[x,\, -y,\, \frac{1}{2}-z\right]$	[18]
19	$\left[-x,\, y,\, \frac{1}{2}-z\right]$	[19]
20	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	[20]
21	$\left[x,\, y+\frac{1}{2},\, z\right]$	[21]
22	$\left[x,\, \frac{1}{2}-y,\, -z\right]$	[22]
23	$\left[-x,\, y+\frac{1}{2},\, -z\right]$	[23]
24	$\left[-x,\, \frac{1}{2}-y,\, z\right]$	[24]
25	$\left[x+\frac{1}{2},\, y,\, z\right]$	[25]
26	$\left[x+\frac{1}{2},\, -y,\, -z\right]$	[26]
27	$\left[\frac{1}{2}-x,\, y,\, -z\right]$	[27]
28	$\left[\frac{1}{2}-x,\, -y,\, z\right]$	[28]
29	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[29]
30	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[30]
31	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[31]
32	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[32]